Software Project Management

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1

A performance management setup for a cloud platform focuses on monitoring, controlling, and improving system performance. The system uses monitoring tools to track CPU usage, memory, storage, and network performance in real time. If performance drops, alerts are sent to administrators so they can act quickly. Load balancing is used to distribute traffic across servers to avoid overload. Auto-scaling is also used to increase or decrease resources depending on demand. Logs are collected and analyzed to find problems and improve performance over time. This setup ensures that the system runs smoothly, users experience fast response times, and downtime is reduced. It helps organizations provide reliable services and maintain customer satisfaction.

2

DiffServ, also known as Differentiated Services, is a network management method used to control traffic and improve quality of service. It works by marking packets with different priority levels. Important data like video calls or online gaming is given higher priority, while less important traffic gets lower priority. Routers read these markings and decide how to handle the data. This helps reduce delays, packet loss, and network congestion. DiffServ is useful in large networks where many types of data are transmitted at the same time. It improves performance by making sure critical services get better network resources.

3

The tc tool in Linux is used to manage and control network traffic. It allows users to set limits on bandwidth, control data flow, and prioritize certain types of traffic. For example, an administrator can limit how much bandwidth a user or application can use. This helps prevent network congestion and ensures fair usage. Using tc requires knowledge of commands and configuration, but it is very powerful. It is useful in managing network performance and improving stability. Overall, tc helps maintain a balanced and efficient network system.

4

A performance management policy defines how a system's performance is monitored and improved. It includes rules for tracking system resources like CPU, memory, and network usage. The policy also defines acceptable performance levels and actions to take when limits are exceeded. For example, alerts can be triggered when usage is too high. It may also include strategies like load balancing and scaling. The purpose of this policy is to ensure the system runs efficiently and reliably. It helps prevent failures, improves user experience, and supports smooth operation of IT services.